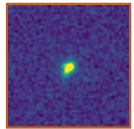
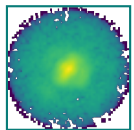


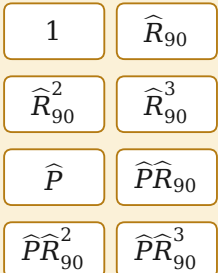
galaxy



PSF

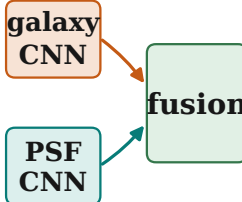


Joint D_4 orbit

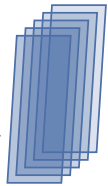


galaxy and PSF transformed jointly

F
one parameter set for all 8 variants



$\phi_0, \dots, \phi_7$



inverse-align
 $\psi_i^{-1}\phi_i$

$$\Psi_{1,2} = \frac{1}{8} \sum_i w_i \psi_i^{-1} \phi_i$$
$$\frac{1}{8} \sum_i \psi_i^{-1} \phi_i$$

signed / sign-free averages

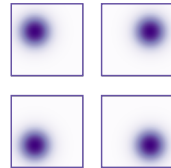


Ψ_2



Attention maps

maps 1-4



from scalar map

bias-free odd MLP

e'_1, e'_2

invariant MLP

size, flux

1. Inputs

53×53 pixels

2. Joint orbit

8 joint variants

3. Shared F

shared weights

Features ϕ_i

$8 \times 13^2 \times 64$

4. Reynolds

signed / invariant

Ψ maps

$3 \times 13^2 \times 64$

5. Pooling

4 spatial maps

6. Output heads

shape + scalars